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ASSIGNMENT -1

Subject : Computer Network

Subject Code :24CSH-337

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Q 2) What is OSI model? What are the different layers of OSI Model? Write Their functionality, Data Format and Protocols used at each layer.

Ans) OSI Model provides a structured layer architecture through which Network Communications are maintained. Each Layer has a specific job and communicates to a layer immediately above it or below it. OSI stands for Open System Intercommunications.

It's architecture consist of 7 layers:

- 1) Application Layer:
 - 1.1) Functionality: Provides network service directly to applications
 - 1.2) Data Format: Data / Message along with appropriate protocol headers.
 - 1.3) Protocol: It can select from various protocols, FTP, HTTPS, SMTP, SFTP and many more
- 2) Presentations Layer:
 - 2.1)1. Functionality: Data Compression, Data Encryption, and Data Translation
 - 2.1)2. Data Format: It doesn't have any specific header.
 - 2.1)3. Protocol: It can have select from various protocols, most appropriate for the job. Like JPEG, PNG, UTF-8, GIF etc
- 3) Session Layer:
 - 3.1) Functionality: Session establishment/Management and Session Termination
 - 3.2) Data Format: No universeral header, only session details like ID, token and etc
- 4) Transport Layer:
 - 4.1) Functionality: End-to-End Communication.
 - 4.2) Data Format: UDP or TCP Header.
 - 4.3) Protocol: it uses either TCP or UDP protocols.
- 5) Network Layer:
 - 5.1) Functionality: Application – to – Application Communication
 - 5.2) Data Format: Protocol Specific Header
 - 5.3) Protocols: IPv4, IPv6, ICMP, IGMP, RIP
- 6) Data Link Layer:
 - 6.1) Functionality: Node-to-Node communication, Error correction and more
 - 6.2) Data Format: It uses divides data into Frame and has a header and trailer.
- 7) Physical Layer:
 - 7.1) Functionality: Actually transmitting the data (raw 0s and 1s) over communication medium whether guided or unguided.

7.2) Data Format: Binary, Raw 0s and 1s either in electrical signals in guided or aptitude and frequency in unguided medium.

Q 2) What are the different Error Detection methods, Explain each with the help of proper examples

A) Error Detections Methods are used to detect that an error or bit flip has occurred due to noise during transmission, with varying degree of confidence of varying amount of n – bits error

1. Parity Check: The network first decided whether to use Even or Odd parity check. In either of those parity checks, the parity of number of 1's in the data should be of the parity chosen on the network that is Odd for Odd parity and Even for Even parity. If not, then parity bit is 1 otherwise 0

For example. DATA: 0 1 1 0 Parity Check: Even

N = 2 (number of 1's in the data)

N is even hence Parity Bit is 0

2. Checksum: The data frame is segmented into 8 or bits block and then add all the blocks together and the 1's Complement of the Result is Checksum bits, Which is appended at the header. The senders adds the checksum to it's result and if all the bits are 1's then there is no error.

For ex. Data block sum : 1 0 0 1

1's Complement: 0 1 1 0

Checksum : 0 1 1 0

3. CRC (Cyclic Redundancy Check): it is more robust algebraic error detection method utilising Binary Division. The sender and receiver agree on a divisor and then sender adds zeros to the data and divides the data with this divisor and leftover remainder is treated as CRC bits appended at the end of the data bits as the trailer. The receiver divides the received message with same divisor and if remainder is 0 then no error occurred.

For ex: Data : 0 1 1 , Divisor: 1 1

Number of Zero's to add $(n-1) = 2$

Quotient : 0 1 1 0 0

The Final Remainder: 1 0 hence the CRC bits are 0.